

Russell Dudek

AI Transformation Operator · Client Advisor · Full-Lifecycle
Builder

Pittsburgh, Pennsylvania
github.com/russelldudek
russelldudek.github.io/CapitalizeAnalytics/

Operator-technologist who makes complex work visible, connects technical and business teams, and builds operating mechanisms that turn emerging technology into adopted, measurable outcomes. Brings AI advisory, agentic workflow design, direct operating transformation, high-reliability engineering, and customer-backward operations leadership.

ROLE-ALIGNED CAPABILITIES

- AI opportunity framing
- Executive discovery
- Technical translation
- Agentic workflow architecture
- RAG & human-in-the-loop patterns
- Governance & escalation design
- Solution scoping & handoffs
- Cross-functional transformation
- Adoption, training & standard work

TECHNICAL TOOLKIT

- Python
- SQL
- JavaScript
- TensorFlow
- LLM workflows
- RAG patterns
- Agentic AI
- Tableau
- ERP / Odoo
- Jira · Confluence
- HubSpot · Slack
- GIS & mapping

CREDENTIALS

- IBM Enterprise Design Thinking Practitioner
- Six Sigma Certification
- OSHA 10

Role thesis: Preserve one proof chain from executive decision through AI mechanism, human authority, evidence, handoff, and client ownership.

SELECTED EXPERIENCE

Vape-Jet 2025–Present

Director of Operations

- Lead AI-first operating transformation across production, purchasing, quality, customer success, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry, and AI-assisted workflows.
- Connect factory, field, customer, and machine-fleet signals into prioritized work, earlier risk detection, cleaner handoffs, dashboards, standard work, training, and accountability.
- Work in a software-enabled robotics and machine-vision environment with approximately 1,000 fielded robots informing support and operating feedback loops.

DudeWorth 2017–Present

Founder / AI and Automation Advisor

- Develop AI-readiness and opportunity frameworks that assess value, feasibility, data readiness, governance, adoption burden, reuse, and time-to-value.
- Design agentic workflow patterns using structured context, retrieval, AI agents, human judgment, escalation, decision records, evaluation, and governance rails.
- Advise on AI-enabled operations, logistics technology, warehouse automation, AMRs, ASRS, cobots, transportation systems, network analysis, and workflow design.

Amazon 2014–2018

Operations Management / Site Leader

- Led launch and execution for Amazon Prime Pittsburgh in a 24/7 customer-backward environment supporting approximately five million second-day orders annually.
- Built leadership routines, labor planning, throughput visibility, escalation paths, standard work, and operating mechanisms.
- Applied Lean and Gemba practices across safety, quality, throughput, customer experience, and team engagement; contributed to Prime Air research and Amazon Flex geolocation concepts.

Compunetics 2000–2013

Director of Operations and Engineering Management

- Progressed through R&D program management, production management, strategic acquisition, operations management, and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery, and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, root-cause practices, performance measures, and process controls; collaborated with high-consequence customers and partners including DoD, CERN, Intel, and AMD.

CONSULTING EVIDENCE

Client-facing technical communication: published industry articles, delivered technical talks, and participated in the HyperTransport Consortium.

Mechanism building: executive and frontline discovery, operating measures, escalation paths, standard work, training, and cross-functional handoffs.

Russell Dudek

Complementary depth · Autonomy, commercial systems,
public-private work, and technical communication

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Candidate vision:

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The career through-line is translation with consequence: move between technical detail, business decisions, customer needs, operating systems, risk, and adoption without losing accountability in the handoff.

CONSULTING FIT

Executive and frontline discovery
Business-first technology framing
Technical presentations and industry talks
Cross-functional program leadership
High-consequence customer environments
Operational measurement and controls
Distributed-team enablement
Regulated autonomy and safety

EDUCATION

University of Pittsburgh

Bachelor of Science

Academic grounding in materials science, physical chemistry, and biology-related studies

LEADERSHIP & COMMUNITY

IEEE

CMPT and EDS Pittsburgh Chapter Chair, 2012–2016

Enviro Social Capital

Chairman of the Board, 2014–2017

ADDITIONAL EXPERIENCE

Cardinal Building Products

2021–2023

Regional Director

- Built the company's first remote regional operation.
- Modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

ZeusVu

2018–2022

Chief Pilot, Autonomous Systems and Aerial Data Operations

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, and governed execution.

Enviro Social Capital

2014–2017

Chairman of the Board

- Co-founded and led a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation.
- Helped shape and evaluate a \$200 million Social Impact Bond feasibility concept for Pittsburgh sewage and green-infrastructure improvement.

IEEE

2012–2016

CMPT and EDS Pittsburgh Chapter Chair

- Led regional technology programming connecting industry, academia, STEM outreach, electronics, manufacturing, and emerging technology.
- Organized conferences, seminars, business tours, and technical programming.

CONCEPT WORK — CLAIM STRENGTH PRESERVED

Concept, not production ownership: AI-enhanced transportation-management and route-planning concepts; TensorFlow-based aerial inventory and computer-vision concepts; medical drone delivery and chain-of-custody concepts; Core and DudexOS agentic operating-model concepts.

POSITIONING FOR CAPITALIZE

Best fit: engagements where the business problem is ambiguous, the solution crosses functions, and credibility depends on staying close to both the executive decision and implementation details.

OPERATING METHODS

Lean and Gemba observation; quality systems and root cause; dashboards and performance measures; escalation paths; standard work; training; decision records; human-in-the-loop governance.